

Dust Simulations for BICEP/Keck

Where should we start, and how should we extrapolate?

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The goal

Build realistic dust foreground simulations for the BICEP/Keck field.

Use them to test whether foreground complexity can bias the recovered value of r .

What BICEP/Keck needs

Maps

- Independent realizations
- BICEP patch
- $N_{\text{side}} = 512$.

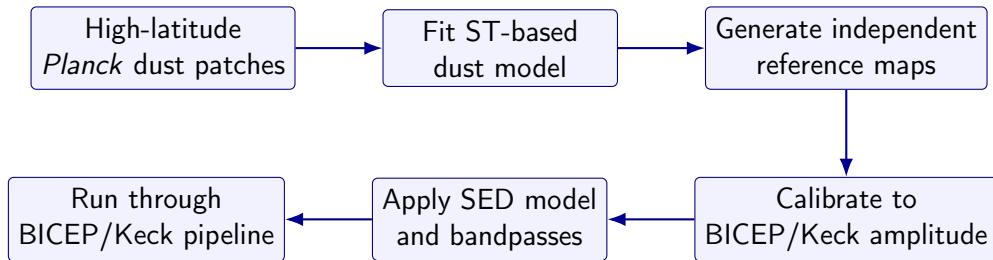
Checks

- Right dust amplitude
- Useful over $30 < \ell < 500$
- Controlled frequency behavior

Not a reconstruction of the real sky: an ensemble of plausible skies.

The range $30 < \ell < 500$ covers the degree scales used by BICEP/Keck, especially the r -sensitive region around $\ell \simeq 80$.

Our proposed route



At what frequency do we build first?

How do we extrapolate?

Mathematical target

We want an ensemble of synthesized foreground maps

$$\left\{ \tilde{\mathbf{s}}_{\nu}^{\prime(a)} \right\}_{a=1}^{N_{\text{sim}}}, \quad \tilde{\mathbf{s}}_{\nu}^{\prime(a)} = \left(\tilde{l}_{\nu}^{\prime(a)}, \tilde{Q}_{\nu}^{\prime(a)}, \tilde{U}_{\nu}^{\prime(a)} \right),$$

on the BICEP/Keck patch and at the BICEP/Keck bandpasses.

In compact form, construct a distribution $\mathcal{P}_{\Theta}$ such that

$$\mathbb{E}_{\mathcal{P}_{\Theta}} \left[\Phi(\tilde{\mathbf{s}}'_{\nu_0}, \tilde{l}') \right] \simeq \hat{\Phi}_{\text{HL}},$$

$$D_{\ell}^{BB,d} \simeq A_{d,\nu_0}^{\text{BK}} \left(\frac{\ell}{80} \right)^{\alpha_d}, \quad 30 \lesssim \ell \lesssim 500.$$

Reference-frequency maps

Starting from high-latitude *Planck* patches, synthesize maps at a reference frequency ν_0 :

$$\tilde{s}_{\nu_0}^{\prime(a)} = \arg \min_{\mathbf{u}} \left[\left\| \Phi_{\text{ST}}(\mathbf{u}, \tilde{l}_{\text{aux}}^{\prime(a)}) - \hat{\Phi}_{\text{HL}} \right\|_2^2 + \lambda_{\text{PS}} \left\| \hat{D}_{\ell}^{BB}(\mathbf{u}) - D_{\ell, \text{target}}^{BB, \nu_0} \right\|_2^2 \right].$$

Then calibrate on the BICEP/Keck mask:

$$\tilde{s}_{\nu_0}^{\prime(a)} \leftarrow \alpha_a \tilde{s}_{\nu_0}^{\prime(a)}, \quad \alpha_a = \left(\frac{A_{d, \nu_0}^{\text{BK}}}{\hat{A}_{d, \nu_0}^{(a)}} \right)^{1/2}.$$

SED model: three cases

Starting from reference-frequency maps, we need a model for $\tilde{Q}'_{\nu}(\hat{n})$, $\tilde{U}'_{\nu}(\hat{n})$ at the BICEP/Keck bandpasses.

1 Spatially varying SED

$$\beta_d(\hat{n}), T_d(\hat{n})$$

2 Multi-layer dust

$$\tilde{Q}'_{\nu}(\hat{n}) = \sum_{k=1}^K \tilde{Q}'_{k,\nu_0}(\hat{n}) f_d(\nu; \beta_{d,k}(\hat{n}), T_{d,k}(\hat{n})).$$

Here k labels distinct dust layers or components along the line of sight.

3 Stochastic SED fluctuations

small-scale SED variations tuned to a target decorrelation level.

Decorrelation diagnostic

For two frequencies, measure on the BICEP/Keck mask:

$$\Delta_d(\nu_1, \nu_2; \ell) = \frac{C_\ell^{BB,d}(\nu_1 \times \nu_2)}{\sqrt{C_\ell^{BB,d}(\nu_1 \times \nu_1) C_\ell^{BB,d}(\nu_2 \times \nu_2)}}.$$

Compare Δ_d with the BICEP/Keck decorrelation constraint.

BK18 peaks at no detected decorrelation; strongly decorrelated models should be treated as labelled stress tests.

Choose an input r and generate a CMB realization.

Add n and one dust realization:

$$\text{CMB}(r_{\text{input}}) + n^{(a)} + \tilde{s}'_{\nu}{}^{(a)}.$$

Run the BICEP/Keck likelihood and measure how dust changes the recovered value:

$$\Delta r = \hat{r}_{\text{recovered}} - r_{\text{input}}.$$

Current ST synthesis script

The script `hlst.py` is our current debugging path for the reference-frequency dust synthesis.

- Start from component-separated *Planck* patch products:

$$(Q_i, U_i), \quad i \in \mathcal{I}_{\text{HL}}.$$

- Compute ST targets separately for Q and U :

$$\hat{\Phi}_Q = \frac{1}{|\mathcal{I}_{\text{HL}}|} \sum_{i \in \mathcal{I}_{\text{HL}}} \Phi_{\text{ST}}(w_i^Q Q_i), \quad \hat{\Phi}_U = \frac{1}{|\mathcal{I}_{\text{HL}}|} \sum_{i \in \mathcal{I}_{\text{HL}}} \Phi_{\text{ST}}(w_i^U U_i).$$

$$w_i^Q = \frac{\sigma_Q(178)}{\sigma_Q(i)}, \quad w_i^U = \frac{\sigma_U(178)}{\sigma_U(i)}.$$

For debugging we can set $w_i^Q = w_i^U = 1$.

- Synthesize one Q map and one U map matching these average targets.

What is fixed, what is adjustable

Fixed for the target

- Target patch STs use non-periodic boundaries.
- ST coefficients are normalized.
- Q and U are treated independently.

Debug knobs

- Patch list, currently patch 129.
- Optional map-amplitude normalization to patch 178.
- Periodic or non-periodic synthesis.

The output is a compact diagnostic: synthesized Q , U , P , and the two loss curves.

Choose a physically sensible extrapolation strategy.

Identify the minimal SED complexity we need.

Help define what “compatible with dust data” means.

Decisions to leave with

- Reference frequency for the first maps.
- SED prescription for the first extrapolation.
- How to allow or limit decorrelation.
- Diagnostics to send with the maps.